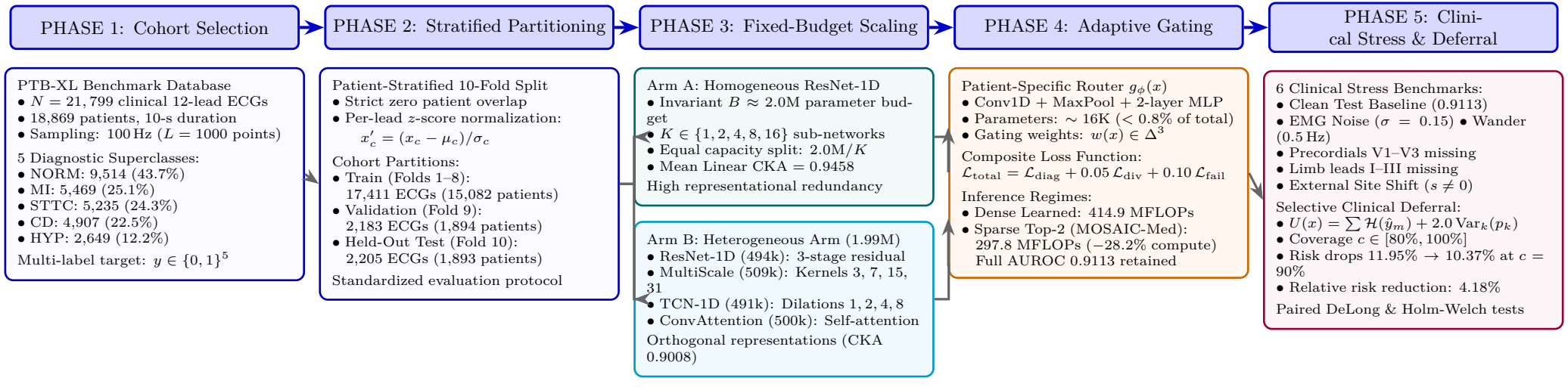
PHASE 1: Cohort Selection
PHASE 2: Stratified Partitioning
PHASE 3: Fixed-Budget Scaling
PHASE 4: Adaptive Gating
PHASE 5: Clinical Stress & Deferral
PTB-XL Benchmark Database
• $N = 21,799$ clinical 12-lead ECGs
• 18,869 patients, 10-s duration
• Sampling: 100 Hz ($L = 1000$ points)
5 Diagnostic Superclasses:
• NORM: 9,514 (43.7%)
• MI: 5,469 (25.1%)
• STTC: 5,235 (24.3%)
• CD: 4,907 (22.5%)
• HYP: 2,649 (12.2%)
Multi-label target: $y \in \{0,1\}^5$
Patient-Stratified 10-Fold Split
• Strict zero patient overlap
• Per-lead $z$-score normalization:
$x'_c = (x_c - \mu_c)/\sigma_c$
Cohort Partitions:
• Train (Folds 1–8):
17,411 ECGs (15,082 patients)
• Validation (Fold 9):
2,183 ECGs (1,894 patients)
• Held-Out Test (Fold 10):
2,205 ECGs (1,893 patients)
Standardized evaluation protocol
Arm A: Homogeneous ResNet-1D
• Invariant $B \approx 2.0$M parameter budget
• $K \in \{1,2,4,8,16\}$ sub-networks
• Equal capacity split: $2.0\text{M}/K$
• Mean Linear CKA = 0.9458
High representational redundancy
Arm B: Heterogeneous Arm (1.99M)
• ResNet-1D (494k): 3-stage residual
• MultiScale (509k): Kernels 3, 7, 15, 31
• TCN-1D (491k): Dilations 1, 2, 4, 8
• ConvAttention (500k): Self-attention
Orthogonal representations (CKA 0.9008)
Patient-Specific Router $g_\phi(x)$
• Conv1D + MaxPool + 2-layer MLP
• Parameters: $\sim$ 16K ($< 0.8\%$ of total)
• Gating weights: $w(x) \in \Delta^3$
Composite Loss Function:
$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{diag}} + 0.05\,\mathcal{L}_{\text{div}} + 0.10\,\mathcal{L}_{\text{fail}}$
Inference Regimes:
• Dense Learned: 414.9 MFLOPs
• Sparse Top-2 (MOSAIC-Med):
297.8 MFLOPs ($-28.2\%$ compute)
Full AUROC 0.9113 retained
6 Clinical Stress Benchmarks:
• Clean Test Baseline (0.9113)
• EMG Noise ($\sigma = 0.15$) • Wander (0.5 Hz)
• Precordials V1–V3 missing
• Limb leads I–III missing
• External Site Shift ($s \neq 0$)
Selective Clinical Deferral:
• $U(x) = \sum \mathcal{H}(\hat{y}_m) + 2.0\,\text{Var}_k(p_k)$
• Coverage $c \in [80\%, 100\%]$
• Risk drops 11.95% $\rightarrow$ 10.37% at $c = 90\%$
• Relative risk reduction: 4.18%
Paired DeLong & Holm-Welch tests